

**Department of Decision Sciences
DNSC 6330: Responsible Machine Learning**

Instructor: Michael Akinwumi, Adjunct Professor

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Class Time & Location: From 03/16/26 - 05/01/26

- Responsible Machine Learning - 48370: Fungler Hall 221, Mon 12:45PM - 03:15PM
- Responsible Machine Learning - 43053: Fungler Hall 208, Tue 04:30PM - 07:00PM

Office Hours: Fungler Hall 415R, Fri, 5:00PM - 7:00PM

Pre-requisite course: DNSC 6314 and 6315, MSBA program candidacy or instructor approval.

Class website: Class materials will be available at this [GitHub page](#). Students will also be directed to additional resources at [Patrick Hall's page](#).

Course Description

Artificial intelligence systems increasingly shape credit decisions, hiring pipelines, health risk scoring, insurance underwriting, and public policy implementation. As these systems scale, so do their legal, operational, and reputational risks. This course prepares future business leaders, model developers, risk managers, and governance professionals to design, evaluate, and oversee machine learning systems with rigor, accountability, and defensibility.

Responsible Machine Learning examines machine learning as a sociotechnical system embedded within institutional, regulatory, and market environments. Students move beyond high-level ethical principles to operational practice: measuring bias, diagnosing model behavior, stress-testing robustness, identifying security vulnerabilities, and designing post-deployment monitoring frameworks.

This course treats Responsible Machine Learning as a discipline of stewardship. Numeric grades are assigned for administrative purposes, but mastery is defined by audit-level rigor, sound reasoning, and the ability to justify decisions under legal, ethical, and operational scrutiny. Students are prepared not only to build models, but to defend them. The result is a practitioner-ready foundation for roles in AI governance, model risk management, ML audit and assurance, and responsible AI leadership within business and regulatory environments.

Course Scope

Topic #	Topic Title	Short Description	Required Reading
1	Foundations of the Alignment Problem	Introduces the alignment problem through real world AI and ML failures, establishes why ML systems are sociotechnical in nature, clarifies the role of Responsible ML and management, and situates the field within emerging Responsible AI career pathways. Responsible Machine Learning will be framed as a risk-benefit framework throughout the course.	a) The Alignment Problem - Machine Learning and Human Values by Brian Christian (2020), Chapters 1 - 3 b) Responsible AI in Practice: A Practical Guide to Safe and Human AI by Toju Duke and Paolo Giudici (2025), Chapter 1 c) Concrete Problems in AI Safety by Dario Amodei et al, https://arxiv.org/abs/1606.06565
2	Explainability and Interpretability as Diagnostic Tools	Examines explainability and interpretability as diagnostic practices rather than a checkbox, comparing model level and system level explanations, assessing tools such as LIME, SHAP, and counterfactuals, and analyzing cases where explainability failed in production across stakeholder contexts.	a) Machine Learning for High-Risk Applications by Patrick Hall, James Curtis, and Parul Pandey (2023), Chapter 2 and Chapter 6 b) Interpretable Machine Learning A Guide for Making Black Box Models Explainable by Christoph Molnar, Chapters 2 - 4, https://christophm.github.io/interpretable-ml-book/
3	Algorithmic Bias Measurement	Focuses on identifying and measuring algorithmic bias, including allocational, representational, and quality of service harms, with grounding in U.S. anti-discrimination laws, fairness definitions, proxy discrimination, intersectionality, and hands-on subgroup evaluation using real datasets.	a) Fairness and Machine Learning: Limitations and Opportunities by Solon Barocas, Moritz Hardt, and Arvind Narayanan (2023), Chapters 6 - 7, https://fairmlbook.org/ b) New Jersey's Disparate Impact Discrimination Claim Analysis; Rules Pertaining to Disparate Impact Discrimination; Guidance on Algorithmic Discrimination; and Introduction to Algorithmic Fairness.
4	Robustness, Generalization, and Dataset Drift	Explores model robustness and failure modes over time, including spurious correlations, stress testing, concept drift versus data drift, and long horizon failures, with applied testing in domains such as credit scoring.	a) Machine Learning Evaluation Towards Reliable and Responsible AI by Nathalie Japkowicz and Zois Boukouvalas (2024), Chapters 11-12

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			b) Concrete Problems in AI Safety by Dario Amodei et al, https://arxiv.org/abs/1606.06565
5	ML Security and Abuse Pathways	Analyzes security threats and misuse of ML systems, including adversarial examples, data poisoning, model extraction, and abuse of generative models, supported by practical exercises in ML security testing.	a) Joseph AD, Nelson B, Rubinstein BIP, Tygar JD. Adversarial Machine Learning. Cambridge University Press; 2019. b) Machine Learning for High-Risk Applications by Patrick Hall, James Curtis, and Parul Pandey (2023), Chapter 5
6	Responsible Deployment and Monitoring	Covers post-deployment responsibilities such as data cards, model cards and system cards, continuous evaluation, alerting and rollback mechanisms, human-in-the-loop design, and organizational roles required for responsible deployment and monitoring.	a) NIST AI Risk Management Framework (RMF), https://www.nist.gov/itl/ai-risk-management-framework b) NIST AI RMF: Generative Artificial Intelligence Profile, https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf
7	Responsible AI Management and Global Regulatory Frameworks	Addresses system-level AI risks associated with foundation and language models, reviews global governance frameworks such as NIST AI RMF, EU AI Act, and ISO/IEC standards, and applies policy analysis through hands-on stakeholder and regulatory gap discussions.	a) Introductory Guide to Responsible AI Management by Denniss Hirsch (2025), https://moritzlaw.osu.edu/sites/default/files/2025-09/Introductory%20Guide%20to%20Responsible%20AI%20Management_0909.pdf b) The OECD AI Principles, https://www.oecd.org/en/topics/sub-issues/ai-principles.html
Capstone	Responsible ML System Audit	Requires students to conduct an end-to-end audit of an ML system of their choice, identify risks and benefits, propose mitigation strategies, and present findings as if before an internal review board or a regulator.	

Reading and Lecture Quizzes

Class participation will be measured by weekly quizzes on the reading and lecture.

Reading Assignments

Students are responsible for studying and understanding all assigned materials *before* class. If reading generates questions that are not discussed in class, the student has the responsibility of addressing the instructor privately or raising the issue in an appropriate digital medium.

Learning Outcomes

As a result of completing this course, students will:

- Diagnose alignment failures in machine learning systems.
- Measure and interpret algorithmic bias using appropriate statistical tools.
- Design robustness, monitoring, and model risk management frameworks.
- Identify security and abuse risks in deployed ML systems.
- Translate Responsible AI principles into enterprise governance structures.
- Communicate technical risk assessments to executive and regulatory audiences.
- Familiarize themselves with important machine learning communities and technologies like GitHub.

Upskilling & Workforce Alignment

Skills you will graduate with include:

- Fairness and bias evaluation
- Explainability tooling
- Model risk assessment
- Responsible deployment practices
- Governance and compliance literacy

Career paths this course supports include:

- Responsible AI Engineer
- AI Governance Analyst
- ML Risk & Compliance Lead
- AI Safety Researcher
- Trustworthy AI Consultant

Additional Assignments

- **Weekly Assignments:** As this will be a 7 week, workshop based course, students are expected to make weekly progress toward a capstone project. Most weeks the instructor will provide an assignment structured to provide the tools and knowledge necessary for the capstone project.
- **Project Performance:** Lecture materials and hands-on workshop materials will be geared toward application to the capstone project. Students are expected to participate in this project as individuals or in groups and to do reasonably well.

- **Public Github Contributions:** Students are expected to write code and generate other artifacts (i.e. notebooks, visualizations, markdown) and to store them in a publicly accessible GitHub repository (or other public location, i.e., personal website, Colab).

Capstone Project: Responsible ML System Audit

Students must:

- Evaluate an ML system of their choice end-to-end
- Identify risks and benefits of the system
- Propose strategies for mitigating the risks
- Present findings as if to an internal review board or regulator

Grading Contributions

Weekly Assignments	25%
Public Github with working codebase	25%
Weekly Quizzes	20%
Capstone	30%

Numeric Grade Scale

Numeric grade	Letter grade	Mastery Band	Meaning
≥ 90.00	A	Distinction	Professional-grade reasoning suitable for internal review boards or regulators
80-89.99...	A-		
75-79.99...	B+	Competent	Solid technical understanding with some gaps in judgment or synthesis
73-74.99...	B		
70-72.99...	B-	Developing	Incomplete integration of technical, ethical, and governance considerations
55-69.99...	C+		
53-54.99...	C		
50-52.99...	C-	Insufficient	Fails to meet graduate-level expectations
≤ 49.99...	F		

A student **cannot** earn an A-range grade without demonstrating professional-level competence on the capstone project.

Generative AI Use Policy

In this course, generative artificial intelligence (GenAI) tools may be used as learning aids for brainstorming, preparatory research, or exploratory analysis **but must always be applied with integrity and transparency**. Consistent with GW's [AI Guidance and Best Practices](#), students must use only AI tools that meet institutional standards for security and data protection, ensure that no restricted or regulated data is input into third-party systems, and critically review and validate all AI-generated content for **accuracy and relevance before incorporating it into submissions**. Any use of GenAI in work submitted for evaluation must be **clearly disclosed with a statement identifying the tool used and the manner of its contribution**. Work presented for grade must be authentically the student's own intellectual product, and reliance on AI to produce evaluative deliverables without permission or appropriate attribution constitutes a violation of academic integrity.

Recommendation Policy

The instructor reserves the discretion to grant (or not) a recommendation request based on:

- Advance notice is required: there must be at least 30 calendar days between the date the instructor receives ALL the necessary materials and the earliest due date of the recommendation form to individual schools.
- Successful completion of the course(s) with strong academic performance (e.g., A-grade, top 10% numeric grade).
- Meaningful class participation and positive peer review.
- Limited to a total of a maximum of 5 recommendation submissions per student

Note that the instructor is **unable** to write visa letters or other immigration supporting documents.

University Policy on Observance of Religious Holidays

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see: <http://provost.gwu.edu/policies-procedures-and-guidelines>.

Academic Integrity

Academic dishonesty is defined as cheating of any kind, including misrepresenting one's own work, taking credit for the work of others without crediting them and without appropriate authorization, and the fabrication of information. For details and complete code, see: <http://studentconduct.gwu.edu/code-academic-integrity>. Violations of the academic integrity code could result in serious disciplinary action.

Contact Hours Statement

This 1.5-credit course includes 2.5 hours of direct instruction (class meetings) and a minimum of 5 hours of independent learning (outside of class), totaling a minimum of 7.5 hours per week, totaling approximately 52 hours for the 7-week term, including exams period.

Disability Support Services (DSS)

Any student who may need accommodation based on the potential impact of a disability should contact Disability Support Services in Rome Hall, 801 22nd Street, NW, Suite 102, to establish eligibility and to coordinate reasonable accommodations. For additional information see: <http://disabilitysupport.gwu.edu>. Phone: 202-994-8250.

Safety and Security

In an emergency: call GWPD 202-994-6111 or 911. For situation-specific actions: review the Emergency Response Handbook: <http://safety.gwu.edu/emergency-response-handbook>. In an active violence situation: Get Out, Hide Out, or Take Out: <http://go.gwu.edu/shooterprep>. Stay informed: <http://safety.gwu.edu/stay-informed>.

Class Policy Changes

The instructor reserves the right to revise any items on this syllabus, including, but not limited to any class policy, course outline or schedule, grading policy, test date, etc. Note that the requirements for deliverables may be clarified and expanded in class, via email, or on Blackboard. Students are expected to complete the deliverables incorporating such additions.